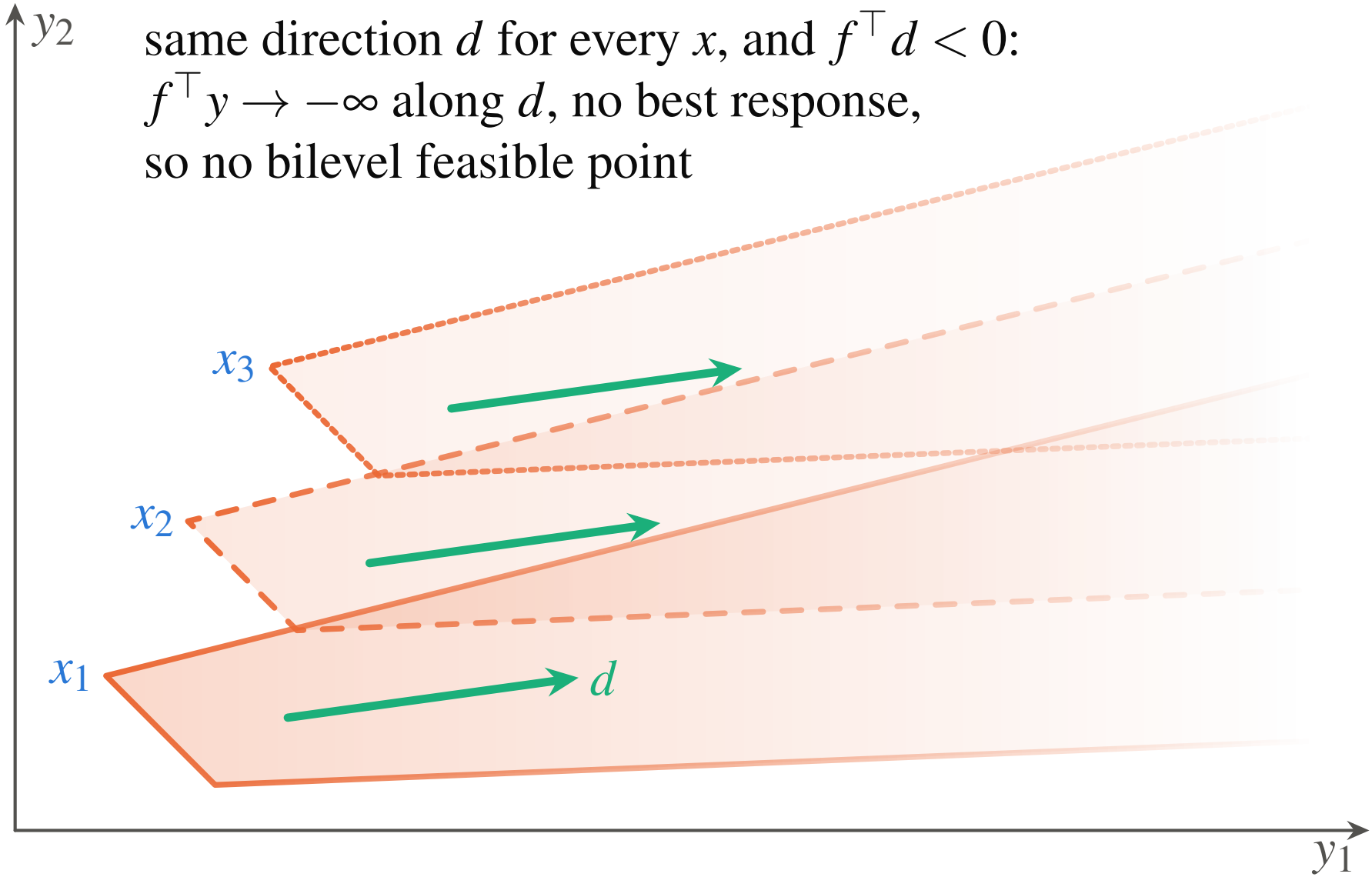
same direction $d$ for every $x$, and $f^\top d < 0$:
$f^\top y \to -\infty$ along $d$, no best response,
so no bilevel feasible point
$y_2$
$x_3$
$x_2$
$x_1$
$d$
$y_1$